

FIRST GROUP MEETING AGENDA

DCIT 204/308 Joint DSA Semester Project

Project Title

Ghana Smart Service Operations Optimizer

1. Meeting Details

Date: _____

Time: _____

Venue/Platform: _____

Meeting Coordinator: _____

Minutes Recorded By: _____

2. Purpose of the Meeting

The purpose of this first meeting is to:

- Allow all 16 members to meet and understand each other's strengths.
- Review the project requirements together.
- Select a suitable Ghanaian context for the project.
- Identify members who are comfortable with coding and members who can support other areas.
- Assign initial responsibilities.
- Agree on communication tools and meeting schedules.
- Set tasks to be completed before the next meeting.

The project is expected to combine custom data structures, algorithms, database storage, testing, performance analysis, graphs, documentation and an oral defence.

3. Meeting Agenda

Agenda 1: Opening and Attendance

Estimated time: 5 minutes

- Welcome all members.
- Record the names and index numbers of everyone present.
- Select someone to record the minutes.
- Select someone to manage attendance.

All members are expected to sign in and out during group meetings.

Agenda 2: Introduction of Members

Estimated time: 15 minutes

Each member should briefly state:

- Full name.
- Index number.
- Programme or level.
- Programming experience.
- Knowledge of Java.
- Knowledge of databases.
- Knowledge of Data Structures and Algorithms.
- Other useful skills.
- Availability for meetings.

Other useful skills may include:

- Report writing.
- Research.
- Data collection.
- Database management.
- Testing.
- Creating graphs.
- Presentation.
- Video editing.
- Project coordination.
- Designing diagrams and flowcharts.

The aim is not to judge anyone. It is to understand how each person can contribute.

Agenda 3: Simplified Review of the Project

Estimated time: 10 minutes

One member should briefly explain the project to the group.

The group is expected to build a Ghana-based service operations system that can:

- Store operational information in a database.
- Load information into custom data structures.
- Prioritise service requests.
- Search and sort records.
- Find routes between locations.
- Assign available resources.
- Test whether algorithms work correctly.
- Measure algorithm performance.
- Present results using tables and graphs.

The project is mainly about correct data structures, algorithms, database integration, testing and performance analysis. It is not mainly a user-interface design project.

Agenda 4: Selection of the Ghanaian Context

Estimated time: 20 minutes

The group should discuss the type of organisation or environment the system will represent.

Possible contexts include:

- University campus service operations.
- Hospital or clinic operations.
- Municipal waste collection.
- Courier or food delivery. (Selected)
- Library or records management. *(Selected)

Each member may suggest an idea. The group should then select one idea based on:

- How easy it is to understand.
- Availability of realistic local information.
- Ability to create enough locations, roads and service requests.
- Ability to apply the required algorithms.
- Ability to complete it within the available time.
- How easy it will be to explain during the oral defence.

Final Selected Context

Selected context: _____

Selected organisation or environment: _____

Proposed project name: _____

Reason for selecting it:

Agenda 5: Define What the System Will Do (Skipped)

Estimated time: 15 minutes

The group should agree on the main problems the system will solve.

Examples include:

1. Determining which service request should be handled next.
2. Finding the shortest route between two locations.
3. Checking which locations are reachable.
4. Assigning an available resource to a request.
5. Searching for a request, location or resource.
6. Selecting requests under a budget, time or capacity limit.
7. Comparing the performance of different algorithms.

The final system should answer operational questions involving queues, priority, routes, connectivity, optimisation and permanent data storage.

Agreed Main System Operations

1. _____
 2. _____
 3. _____
 4. _____
 5. _____
-

Agenda 6: Discuss Members' Skills and Work Areas (Skipped)

Estimated time: 20 minutes

Not every member may be strong in coding. Therefore, responsibilities should be assigned according to individual strengths.

However, all members must understand the general project and should not depend entirely on the stronger programmers.

The work can be divided into the following areas:

A. Coding and Algorithm Team

This team will mainly handle:

- Custom data structures.

- Search and sorting algorithms.
- Graph algorithms.
- Greedy and dynamic programming algorithms.
- System integration.

Members who are more comfortable with Java and coding should lead this area.

B. Database and Dataset Team

This team will handle:

- Database tables.
- CSV files.
- Ghanaian locations and routes.
- Service requests.
- Resources.
- Data validation.
- Connecting the database to the Java program.

C. Testing and Performance Team

This team will handle:

- Unit tests.
- Edge-case testing.
- Trace tables.
- Runtime experiments.
- Recording test results.
- Exporting results to CSV.
- Creating performance graphs.

The project requires at least 40 tests and multiple trace tables across the structures and algorithms.

D. Documentation and Report Team

This team will handle:

- Meeting minutes.
- Development log.
- Problem statement.
- Data dictionary.
- System architecture.
- Flowcharts and diagrams.
- Technical report.
- Individual contribution records.
- Screenshots and appendices.

E. Presentation and Demonstration Team

This team will help with:

- Preparing the demonstration.
- Organising the oral defence.
- Creating presentation materials.
- Preparing the demonstration video.
- Ensuring every member understands their assigned topic.

These teams can overlap. A strong programmer may also help with testing, while a documentation member may also assist with collecting data.

Agenda 7: Individual Learning and Oral Defence Preparation

Estimated time: 10 minutes

Although some members may do more coding than others, every member must be prepared to explain at least:

- One data structure.
- One algorithm.

This is required because every member may be questioned individually during the oral demonstration.

A member who is not confident in coding can work closely with a stronger coder. The stronger coder may handle most of the implementation, while the other member studies:

- What the structure or algorithm does.
- How it works.
- Its inputs and outputs.
- Its advantages and limitations.
- Its time complexity.
- A simple trace example.
- How it is used in the group's system.

Initial Assignment Table

Member	Main Work Area	Data Structure to Study	Algorithm to Study
Member 1			
Member 2			
Member 3			
Member 4			
Member 5			

Member 6

Member 7

Member 8

Member 9

Member 10

Member 11

Member 12

The final assignments do not have to be completed during the first meeting. They can be reviewed after the group understands everyone's strengths.

Agenda 8: Select Group Leaders and Coordinators

Estimated time: 10 minutes

The group should select people to coordinate the following areas:

Position	Name
Group Coordinator	
Assistant Coordinator	
Technical/Coding Lead	
Database and Dataset Lead	
Testing and Performance Lead	
Documentation Lead	
Presentation Lead	
Attendance and Minutes Officer	

The group coordinator should not make every decision alone. Major decisions should be discussed with the team.

Agenda 9: Agree on Tools and Communication

Estimated time: 10 minutes

The group should agree on the tools that will be used.

Suggested Tools

Purpose	Suggested Tool
Group communication	WhatsApp
Online meetings	Google Meet or Zoom
Documentation	Google Docs
File storage	Google Drive
Source-code management	GitHub
Task tracking	Trello, Notion or Google Sheets
Programming	Java
Database	SQLite, MySQL or PostgreSQL
Graph creation	Excel, Python or Java

The brief recommends Java, while SQLite, MySQL or PostgreSQL may be used for the database.

Agreed Tools

Communication platform: _____

Document storage: _____

Source-code platform: _____

Database: _____

Task management tool: _____

Agenda 10: Agree on Group Rules

Estimated time: 10 minutes

The group should agree on simple rules such as:

- Members should attend scheduled meetings.
- Members should inform the group when they cannot attend.
- Every assigned task should have a deadline.
- Members should report difficulties early.
- Stronger members should support weaker members.
- No member should remain without a responsibility.
- Code should be explained before it is accepted.
- Work completed by each member should be recorded.
- Members should not submit copied generic projects.
- Important decisions should be recorded in the meeting minutes.

The project requires individual accountability, even though it is completed as a team.

Agenda 11: Assign Tasks Before the Next Meeting

Estimated time: 10 minutes

The first tasks should focus on understanding and planning the project.

Suggested First Tasks

Task	Assigned Member(s)	Deadline
Write the initial problem statement		
Research the selected Ghanaian context		
Prepare a list of local locations		
Prepare sample service requests		
Draft the database tables		
Prepare the initial data dictionary		
List the required data structures		
List the required algorithms		
Create the GitHub repository		
Create the Google Drive folder		
Prepare meeting minutes		
Prepare the development log		

The first recommended milestone is to complete the problem statement, dataset plan, data dictionary and draft database schema.

Agenda 12: Schedule the Next Meeting

Estimated time: 5 minutes

Next meeting date: _____

Next meeting time: _____

Meeting venue/platform: _____

Main purpose of the next meeting:

4. Expected Outcomes of the First Meeting

Before the meeting ends, the group should have:

- Recorded all members' names and index numbers.
 - Identified members' skills and availability.
 - Selected a Ghanaian context.
 - Chosen a provisional project title.
 - Defined at least five major system operations.
 - Identified the stronger programmers.
 - Identified members who can support the database, testing, documentation and presentation.
 - Selected group coordinators.
 - Agreed on communication and collaboration tools.
 - Assigned initial tasks.
 - Set deadlines.
 - Scheduled the next meeting.
 - Recorded attendance and meeting minutes.
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5. Important Note to Members

The project should not be treated as work for only the members who can code.

The stronger programmers may handle more of the difficult implementation, but other members should contribute through:

- Dataset preparation.
- Database design.
- Testing.
- Trace tables.
- Performance experiments.
- Graph creation.
- Research.
- Report writing.
- Diagrams.
- Presentation.
- Demonstration preparation.

Every member should still understand the project and prepare to explain an assigned data structure and algorithm during the oral defence. The final submission must also include an individual contribution statement.

FIRST MEETING MINUTES

Date: _____

Time started: _____

Time ended: _____

Venue: _____

Chairperson: _____

Minutes recorded by: _____

Members Present

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____

Members Absent

Main Decisions Made

1. _____
2. _____
3. _____
4. _____
5. _____

Tasks Assigned

Task	Member Responsible	Deadline
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Challenges or Concerns Raised

Date of Next Meeting

Signatures

Group Coordinator: _____

Minutes Secretary: _____